A bright spark

At first, Michael worked on projects involving combustion (burning) and explosive chemicals. However, he soon became captivated by a new discovery in physics: when electricity flows through a metal wire, it turns the wire into a weak magnet.

Alongside his work for Humphrey, Michael quietly started his own research. In 1821, he figured out how to arrange a wire and magnet so that when electricity passed through the wire, it began swinging around the magnet in a circle. He had invented the electric motor—changing electricity into movement for the first time in history.

Lighthouses once used oil lamps and reflective surfaces for light. Michael spent years looking for a way to use a generator instead. Finally, in 1858, electric lighting was installed in a working lighthouse for the first time, with Michael's approval.

Electric light first shone out to sea from the South Foreland Lighthouse in Dover, UK.

Michael gave amazing science lectures at the Royal Institution in London to inspire children just as he had been inspired by Humphrey Davy.

In 1822, Michael scribbled a reminder in his calendar to try the experiment in reverse. Ten years later, he finally got around to the task—and showed that it was possible to generate electricity using magnets and movement. Now there was a way to produce electricity without needing to use batteries, which can run out. Michael's ideas and discoveries sparked an age of invention and experimentation!

Michael helped Humphrey Davy invent a lamp that was safe to use in coal mines. Before that, lamps often caused explosions.